

Orchestration Flow

Introduction

Orchestration is the process of coordinating all components of the AI evaluation system to ensure a smooth and efficient workflow. It defines how data flows between the user, Retrieval-Augmented Generation (RAG) pipeline, evaluation agents, and the final verdict generation module. The orchestration layer ensures that each component performs its task in the correct sequence and that the final response is accurate, reliable, and well-evaluated.

1. Complete Workflow

The overall workflow of the proposed system is as follows:

1. The user submits a query through the web application.
 2. The query is processed by the LangChain RAG pipeline.
 3. The query embedding is generated using the Sentence Transformer model.
 4. FAISS retrieves the most relevant document chunks.
 5. The retrieved context and user query are sent to the Gemini LLM.
 6. Gemini generates a context-aware response.
 7. The generated response is evaluated by all evaluation agents.
 8. RAGAS calculates retrieval and response quality metrics.
 9. TruLens monitors the execution and records performance metrics.
 10. The Verdict Agent combines all evaluation scores and generates the final quality report.
 11. The verified response is displayed to the user.
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2. Agent Communication

The evaluation follows a parallel multi-agent approach.

Each evaluation agent receives the same generated response but evaluates a different quality aspect.

- **Relevance Agent** → Checks whether the response answers the user's query.
- **Accuracy Agent** → Verifies factual correctness.
- **Hallucination Agent** → Detects unsupported or fabricated information.
- **Completeness Agent** → Checks whether all important concepts are covered.
- **Verdict Agent** → Collects scores from all agents and produces the final evaluation.

This parallel communication improves evaluation speed and modularity.

3. Retrieval Flow

The retrieval process ensures that the LLM generates responses using trusted document sources instead of relying only on its internal knowledge.

Retrieval Steps

1. Documents are uploaded and preprocessed.
2. Documents are divided into chunks using Recursive Character Text Chunking.
3. Each chunk is converted into embeddings using Sentence Transformers.
4. Embeddings are stored in the FAISS Vector Database.
5. User queries are converted into embeddings.
6. FAISS performs semantic similarity search.
7. The Top-K relevant document chunks are retrieved.
8. Retrieved context is provided to the Gemini LLM for response generation.

This retrieval mechanism improves factual accuracy and reduces hallucinations.

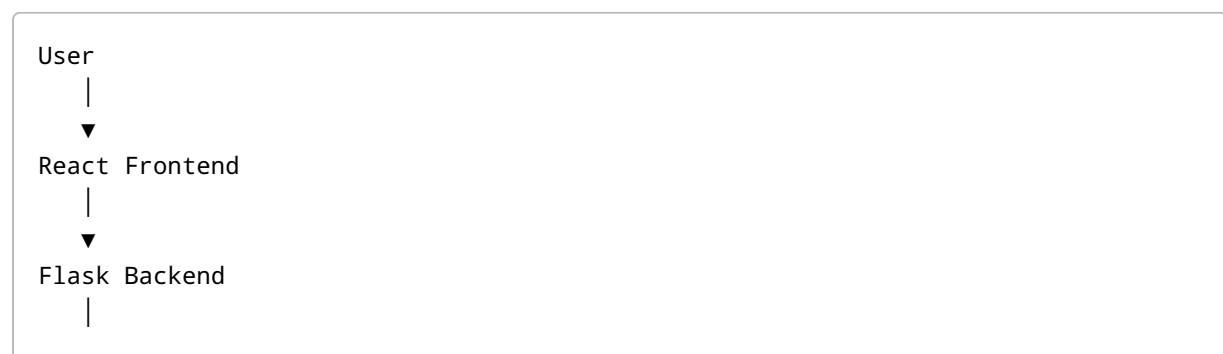
4. Verdict Generation

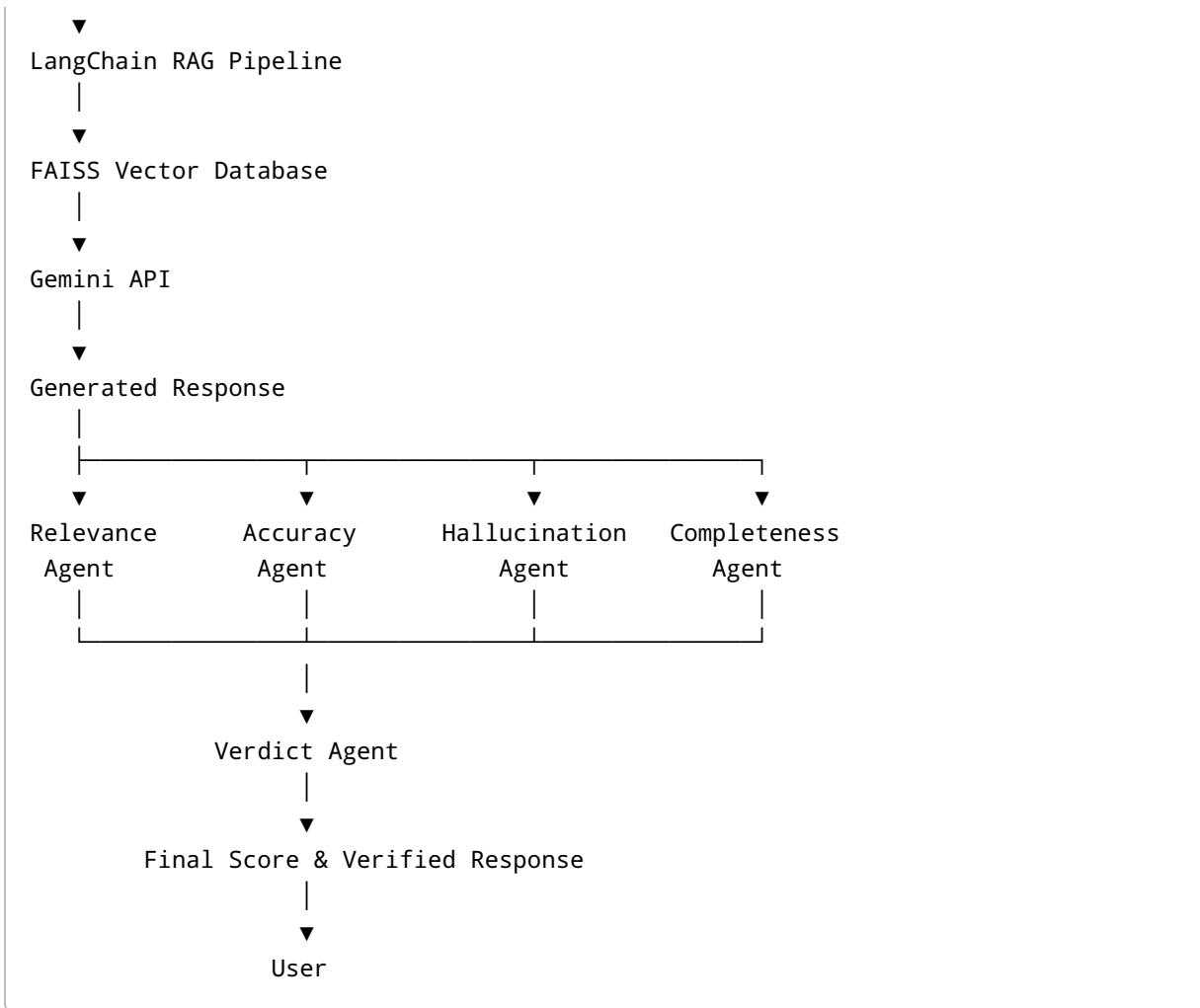
After all evaluation agents complete their analysis, the Verdict Agent generates the final evaluation.

The Verdict Agent:

- Collects scores from all evaluation agents.
 - Calculates the overall quality score using the predefined scoring formula.
 - Assigns a response grade (Excellent, Good, Average, or Poor).
 - Generates a natural language explanation highlighting the strengths and weaknesses of the response.
 - Returns the final verified response along with the evaluation summary to the user.
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Overall Orchestration Flow





Conclusion

The orchestration flow coordinates the complete AI evaluation pipeline, from user query processing and document retrieval to response generation and multi-agent evaluation. By integrating the RAG pipeline, Gemini LLM, RAGAS, TruLens, and the Verdict Agent, the system delivers accurate, explainable, and trustworthy AI responses while ensuring efficient communication between all components.